Why Is Your Electric Bill Going Up? Evidence from Four PJM States

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After adjusting for inflation, residential electricity prices rose 22 to 29 percent in four PJM states between 2005 and 2025. Of the factors examined, capacity prices, what PJM pays power plants to be available during peak demand, showed the clearest relationship with rising residential electricity costs. Data centers are a plausible driver, but no public data show their actual electricity use, so the key step is to require large facilities to report it.

Are data centers pushing up your electric bill, and what can the data actually show?

About 78 percent of Americans worry that data centers are raising electricity prices, and more than 300 data-center bills were introduced in state legislatures in early 2026. Yet the debate runs ahead of the evidence, because no public source reports how much electricity data centers use.

PJM runs the wholesale electricity market for 13 states and D.C., including the four studied here. It pays power plants twice: for the electricity they generate, and for being available when demand is highest. The second payment, the *capacity price*, is set years ahead in auctions based on a *forecast* of future demand, and customers pay for it through their bills.

This study uses 21 years of monthly data (2005 to 2025) for four PJM states to ask which factors move with residential prices. I compare prices across the four states and across months, adjusting for each state's usual price level, the time of year, weather, natural gas prices, wholesale prices, and manufacturing activity. The results show how several supply and demand factors interrelate, but they do not prove what causes prices to rise.

KEY FINDINGS

- **Prices rose in all four states.** Inflation-adjusted prices are up 22 to 29 percent since 2005, led by Ohio at 29 percent.
- **Capacity prices rose along with them.** A $100 higher capacity price is associated with about 0.3 cents more per kWh.
- **The last three auctions cleared at the price ceiling.** Virginia's 2025/26 price was 82 percent above any earlier year.
- **The available data cannot show whether data centers are raising prices.** Virginia's commercial sales grew 55 percent while its residential price rose only 3.6 percent.
- **The key step is transparency.** Requiring large facilities to report electricity use will help to answer the question.

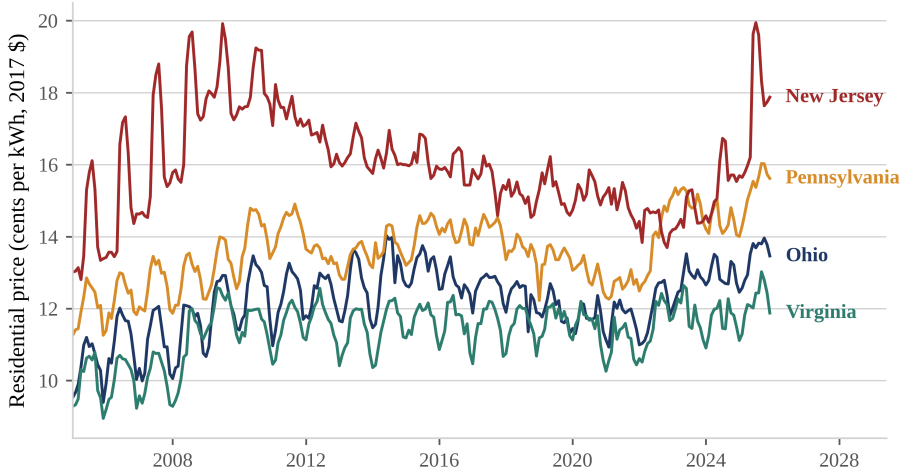
FINDING 1

Residential electricity prices rose 22 to 29 percent in real terms in all four states.

Comparing the first and last twelve months in the sample, prices rose from 10.38 to 13.38 cents per kilowatt-hour in Ohio (29 percent), 12.02 to 15.26 in Pennsylvania (27 percent), 14.03 to 17.50 in New Jersey (25 percent), and 9.96 to 12.12 in Virginia (22 percent). All figures are adjusted for inflation, so the increase is on top of rising prices elsewhere in the economy.

Price levels and paths differ by state. New Jersey's price fell for a decade before its 2025 spike, so these are changes between two points in time, not steady trends.

Real residential prices are higher in every state than in 2005

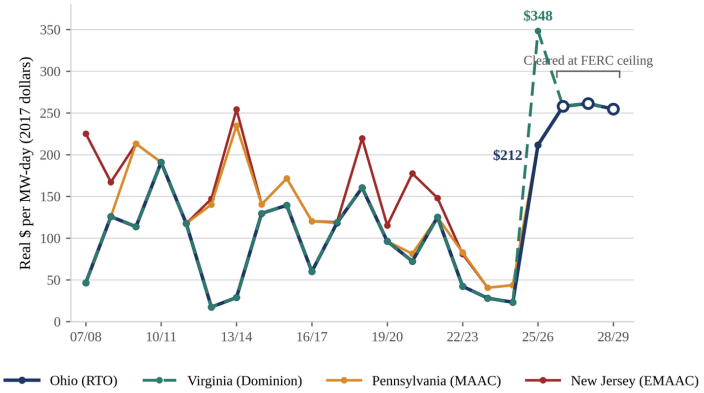


This figure shows monthly average residential electricity prices in cents per kWh, in constant 2017 dollars, for each state from 2005 to 2025. Source: U.S. Energy Information Administration.

FINDING 2

The capacity price is clearly linked to higher prices, and it has jumped.

Capacity prices have hit record highs and the price ceiling



This figure shows the price PJM's auctions paid for capacity, in constant 2017 dollars per megawatt-day, for each delivery year. Open circles are the last three auctions, which cleared at the price ceiling. Source: PJM Interconnection.

Ohio's 2025/26 capacity price of $212 was 11 percent above its previous high, and Virginia's $348 was 82 percent above any earlier year. In the statistical model, a $100 increase in the capacity price (per megawatt-day) is associated with about 0.3 cents more per kilowatt-hour.

Table 1. What moves with residential prices

Factor	Effect (¢/kWh)	p-value
Capacity price ($/MW-day)	**+0.0032**	**0.002**
Natural gas price ($/Mcf)	–0.0070	0.810
Wholesale price ($/MWh)	+0.0005	0.904
Manufacturing GDP ($B)	+0.0115	0.555
Commercial sales (M MWh)	+0.3004	0.173
Information-sector GDP ($B)	–0.0347	<0.001

Effect per one-unit increase in each factor, from 892 monthly observations (June 2007 to December 2025) with state and calendar-month effects. Its negative sign suggests information-sector GDP is a poor data-center proxy.

This shows a relationship, not a cause. By state it is statistically significant only in Pennsylvania and Virginia, and with four states the model cannot rule out other explanations.

FINDING 3

The available data cannot show whether data centers are raising residential prices.

Capacity prices respond to forecasted demand, and forecasts can include announced data centers, so data centers are a plausible driver, but other demand drivers cannot be excluded. No public source reports their actual electricity use by state. Two proxies were used, commercial electricity sales and information-sector GDP, but neither isolates data centers.

Separate research by EPRI finds that data centers already built may have lowered U.S. residential rates by about 6 percent through 2024, because their demand spreads fixed grid costs over more electricity. The same researchers note that this reverses if forecasted demand does not materialize. The two findings are consistent, since realized and forecasted demand can push prices in opposite directions.

WHY VIRGINIA IS HARD TO READ

+55%
Virginia commercial electricity sales, 2019 to 2025, which EIA attributes mainly to data centers

+3.6%
Virginia real residential price over the same period, the smallest of the four states

Virginia is also the only regulated market of the four, so this comparison cannot separate the effect of data centers from the effect of market structure.

There is not sufficient evidence in the current data to find a causal link between data centers and residential prices.

Takeaways for policy

- **Require large facilities to report their actual electricity use** to state regulators, published in aggregate. Utilities already meter it, so the data exist; what is missing is publication. It sets no prices and blocks no construction.
- **Have PJM publish how much forecast demand comes from large loads,** how much has been screened, and compare forecasts with actual load each year.
- **Pair reporting with large-load tariffs,** such as the one Ohio regulators ordered for AEP Ohio in 2025, so the cost of serving data centers stays with the facilities that cause it.
- **Watch the December 2026 auction.** A third consecutive shortfall in reserve capacity would trigger a reliability backstop auction under current rules.

Want to learn more?

Read the full report (PDF): link in the post comments, or click here · Contact: comune.5@osu.edu

Data: EIA, PJM, Bureau of Economic Analysis, NOAA, and Watten, Bistline & Blanford (EPRI, 2026). Monthly data, 2005–2025.